

Definition and first terms

$24 A(x) = 23 + 64 x + A(x)^8$, with the origin-normalized solution.
 Normalization/grammar: $A(x)=1+(4)*T(x)$; $T=x+7*T^2+56*T^3+280*T^4+896*T^5+1792*T^6+2048*T^7+1024*T^8$.
 $a(0), \dots, a(7) = 1, 4, 28, 616, 15820, 453208, 13894552, 445970128$.
 Kernel used throughout: $\rho(u) = u \left(1 - 7 u^1 - 56 u^2 - 280 u^3 - 896 u^4 - 1792 u^5 - 2048 u^6 - 1024 u^7\right)$, so $x = \rho(T(x))$.

Typogeometry

1x4

{1,1}x28

{1,{1,1}}x196

{1,1,1}x224

colored arity geometry; node color distinguishes constructors. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: Δ_2 : 7, Δ_3 : 56, Δ_4 : 280, Δ_5 : 896, Δ_6 : 1792, Δ_7 : 2048, Δ_8 : 1024.

Coefficient and contour extraction

$$\mathcal{K}_n = \{(k_1, \dots, k_7) \in \mathbb{Z}_{\geq 0}^7 : \sum_{j=1}^7 j k_j = n - 1\}, \quad K = \sum_{j=1}^7 k_j,$$

$$a(n) = \frac{4}{n} \sum_{\mathbf{k} \in \mathcal{K}_n} \binom{n + K - 1}{K} \binom{K}{k_1, \dots, k_7} 7^{k_1} 56^{k_2} 280^{k_3} 896^{k_4} 1792^{k_5} 2048^{k_6} 1024^{k_7}.$$

$$a(n) = \frac{4}{2\pi i n} \oint_{\gamma} \frac{du}{\rho(u)^n}, \quad n \geq 1.$$

Recurrence

$\sum_{r=0}^7 P_r(n) a(n+r) = 0$ for $n \geq 1$. The exact degree-7 coefficient polynomials are embedded in `certificate_payload.json` (source: `data/recurrence.json`).

Differential equation

The scalar linear ODE has order 7. It is obtained from the recurrence by the standard Euler-operator substitution. Exact coefficients and boundary polynomial: `data/ode.json`.

Matrices, telescoper certificate, and checks

No compact matrix tuple is shown; see the embedded exact reduction data.

Certificate.

$R(n, u) = N(n, u) / \rho(u)^6$, $\deg_u N = 49$.
 Telescoping identity: $\sum_r P_r(n) H_{n+r}(u) = \partial_u (R(n, u) H_n(u))$.
 Matrix dimensions: 16×16 ; remainder 7×8 , rank 7, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.